



Composable
Commerce Starters

PriceProvider

Composable Starter Template
for Price Management

"Take full control – focus development on your own needs"

Spring Boot

Angular

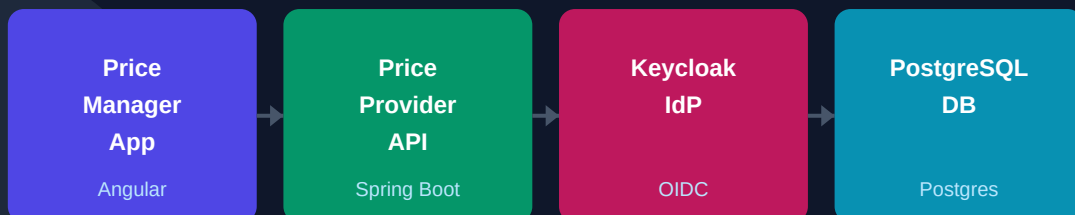
PostgreSQL

Keycloak OIDC

Docker

REST API

- ❖ Multi-currency & quantity-tier pricing
- ❖ RBAC & OIDC organization-scoped access
- ❖ Public headless API – plug any frontend
- ❖ One-command Docker Compose startup
- ❖ AI-agent skills for rapid customization



■ THE COMPOSABLE STARTER TEMPLATE CONCEPT

PriceProvider is a **composable starter template** – fully working, production-ready code that you own and extend. Instead of integrating a black-box SaaS, you get the entire source: every layer from database to REST API to Angular UI. AI agent skills (GitHub Copilot / Claude / GPT) are **pre-configured** to let you add features through natural-language prompts.

Why composable starter templates?



Full Control

You own the source. No vendor lock-in, no black-box pricing engine. Modify the price-selection algorithm, add new price types, or swap the database.



Agentic Engineering

Pre-defined AI agent skills (entity-creation, query-filtering, bulk-ops, RBAC, i18n) let GitHub Copilot Agents extend the service and app with a single prompt.



Headless & Composable

The REST API is completely decoupled from any frontend. Connect your shop system, PIM, ERP, or any custom frontend – or use the included Angular admin app.



Production-Ready Day One

Docker Compose stack, Keycloak OIDC auth, PostgreSQL, sample data, Postman tests and Helm charts for Azure Container Apps are all included out of the box.



Layered Architecture

Clean separation: Data Access → Service → Facade → Controller. Each layer follows Interface-Driven Design so components are independently testable.



Multi-tenant by Design

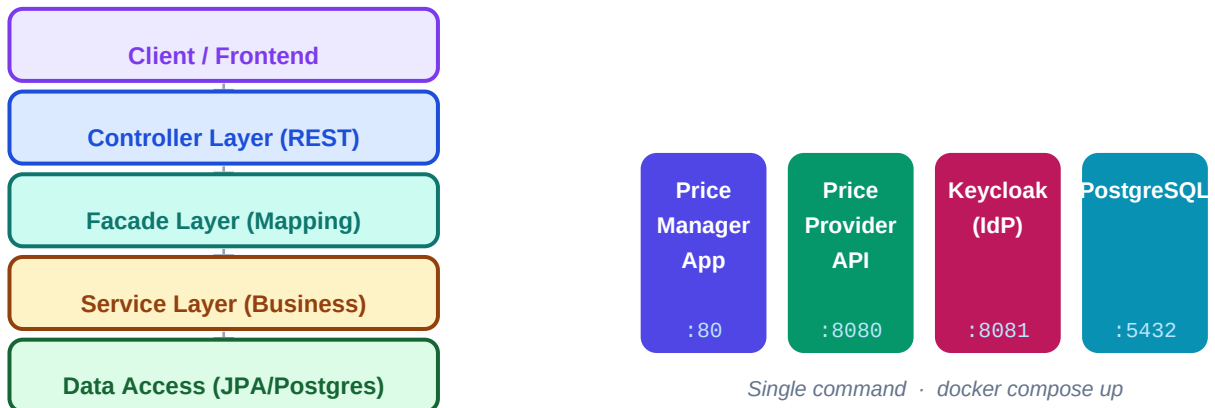
Organizations, channels, countries, currencies, units, tax classes – every dimension is configurable. B2B contract prices, promotions, and customer tiers all fit.

■ FUNCTIONAL FEATURES

ID	Feature	Description
FF-01	Headless REST API	Store, update & query prices via REST
FF-02	Multi-currency	EUR, USD, GBP and any custom currency
FF-03	Net / Gross	Prices stored as net or gross; channel drives display
FF-04	Quantity tiers	Different prices per minimum quantity breakpoint
FF-05	Date validity	validFrom / validTo ranges; newest-wins fallback
FF-06	Customer prices	Organization-scoped via JWT groups claim

FF-07	Promotion groups	Group hierarchy traversal via recursive SQL CTE
FF-08	Bulk upsert	Bulk create-or-update with natural key matching
FF-09	Lucene filtering	q= query parameter on list endpoints
FF-10	Tax calculation	Force-net / Force-gross per channel, tax class support

■■ ARCHITECTURE & TECHNOLOGY STACK



Technology Stack

Java 21

Spring Boot 3

Angular 20

PostgreSQL

Keycloak

Docker

Gradle

Bootstrap 5

Transloco

■■ PRICE MANAGER APP · Angular Admin UI

The **Price Manager App** is a full Angular 20 single-page application that gives admins complete control over every entity: price rows, units, currencies, tax classes, channels, countries, organizations, roles and permissions. The UI is fully internationalized (Transloco) with English and German out of the box.

■ Price Manager

■ Dashboard

■ Price Rows

■ Products

■ Organizations

■ Channels

■ Countries

■ Users/Roles

Price Rows

■ Search price rows...

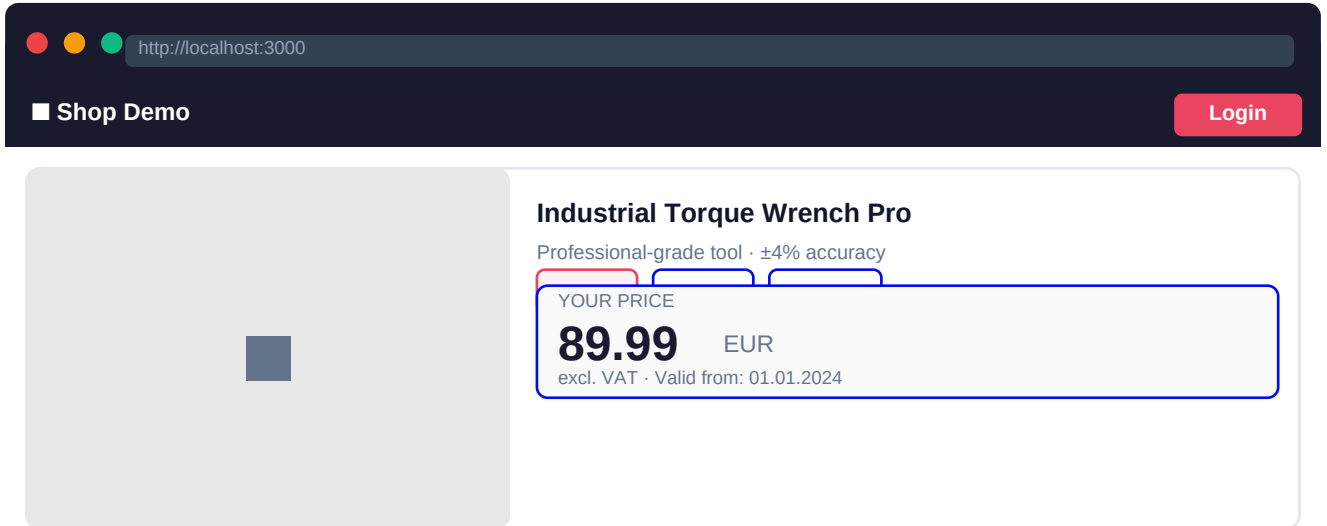
+ New Price

ID	Product	Price	Currency	Min Qty	Valid From	Type
#1	DEMO-PROD-001	99.99	EUR	1	2024-01-01	SALES
#2	DEMO-PROD-002	99.99	EUR	10	2024-01-01	SALES
#3	TOOL-DRILL	25.00	EUR	1	2025-01-01	RENTAL

Admin pages: Dashboard · Price Rows · Units · Currencies · Tax Classes · Channels · Countries · Organizations · Languages · App Roles · App Permissions · Service Initialization

■ EXAMPLE FRONTEND: Shop Demo

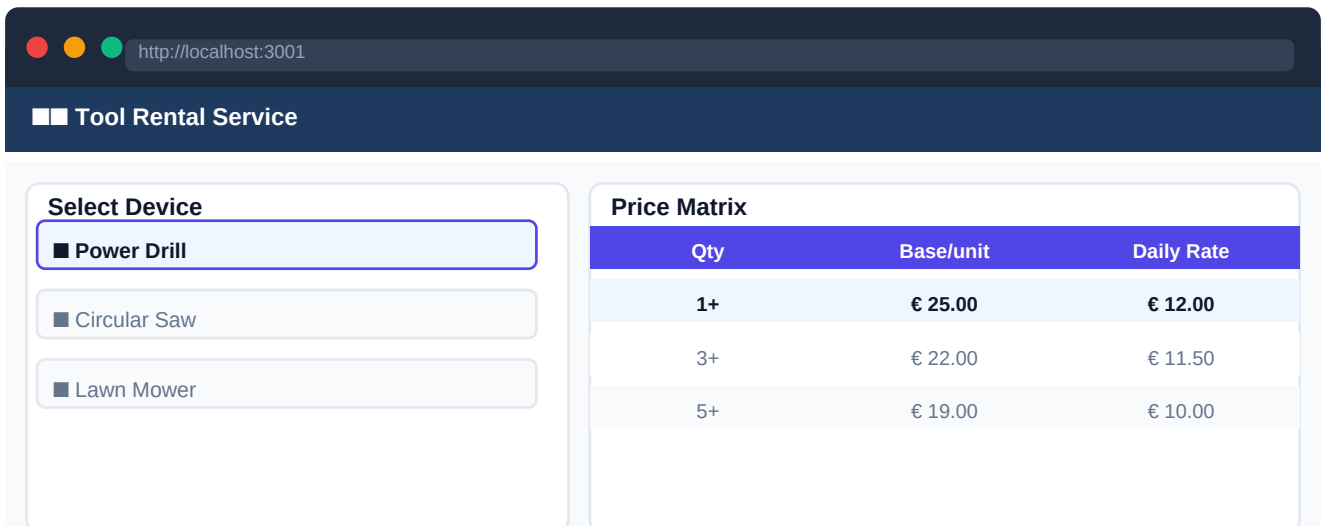
A minimal, ready-to-run shop page that demonstrates the entire integration pattern: anonymous public pricing, OIDC login via Keycloak, and automatic organization-specific contract prices based on the user's groups JWT claim.



localhost:3000 · Vanilla JS + Keycloak adapter · No framework required

■■ EXAMPLE FRONTEND: Tool Rental Demo

Showcases a **two-component rental pricing model**: a one-time base fee per device plus a daily rate. Quantity and duration tiers drive bulk discounts. A live price matrix shows all breakpoints. Organization contracts are applied after Keycloak login.



localhost:3001 · Vanilla JS · Demonstrates bulk & all-quantities API endpoints

■ PUBLIC PRICE API (headless, RESTful)

Any frontend – shop system, ERP, PIM, mobile app – can integrate prices via a single HTTP call. The service selects the best-matching price considering date validity, quantity tier, currency/unit, and the caller's organization (from JWT).

GET

`/public/api/channels/{ch}/countries/{co}/pricedresource/{id}/{type}`
Best matching price for a product SKU

GET

`/public/api/channels/{ch}/countries/{co}/pricedresource/{id}/{type}/`
All matching prices ranked by priority

GET

`/public/api/channels/{ch}/countries/{co}/pricedresource/{id}/{type}/`
All quantity tiers for display in price tables

GET

`/public/api/channels/{ch}/countries/{co}/pricerows/{type}`
Bulk "best price" query for multiple SKUs at once

Price Selection Priority

Priority	Criterion	Rule
1 (highest)	Organization	Prices assigned to the user's JWT organization win
2	Group Distance	Direct org match beats parent group
3	Valid Date	Newer validFrom beats older validFrom
4	Quantity Tier	Higher minQuantity (closer to requested) wins

■ QUICK START · One Command

Prerequisites: Docker Desktop installed and running.

```
./build-and-run-devenv.sh # Linux / macOS  
build-and-run-devenv.bat # Windows
```

Service	URL	Description
Price Manager App	http://localhost	Angular admin UI
Price Provider API	http://localhost:8080	REST API + Swagger UI
Keycloak (IdP)	http://localhost:8081	Identity Provider
Shop Frontend	http://localhost:3000	Demo shop (requires Node.js)
Rental Frontend	http://localhost:3001	Demo rental (requires Node.js)

■ AGENTIC ENGINEERING · AI Agent Skills

PriceProvider ships with **pre-defined GitHub Copilot / Claude agent skills**. Each skill is a detailed instruction file that guides an AI coding agent through a complete, multi-layer feature addition – ensuring architectural consistency without manual scaffolding. *Development focused entirely on your own needs.*



Entity Creation & Update

Add a new domain entity across all layers: JPA entity, repository, service, facade, REST controller, Angular list/detail pages, and i18n strings.



Query Filtering

Add Lucene-like q= filtering to any list endpoint. The skill generates SpecificationBuilder integration, controller params, and Angular filter UI.



Bulk Operations

Implement bulk create-or-update with natural key matching. Supports CSV/JSON ingestion, conflict resolution, and partial-success reporting.



Security & RBAC

Add roles, permissions, and organization-scoped data access. Generates AppRole/AppPermission JSON seeds plus Angular route guards.



Angular Components

Generate standalone Angular components with signals, reactive forms, Bootstrap layout, and Transloco i18n following all project conventions.



Translation

Add or update i18n strings in both backend (Java MessageSource) and frontend (Transloco JSON) for all supported languages simultaneously.

■ SECURITY & RBAC · Keycloak OIDC

Role	Access Level	Typical User
priceprovider.admin:Superuser	All permissions incl. permission management	System owner
priceprovider.admin:Admin	Full read/write/delete on all entities	Administrator
priceprovider.admin:Contributor	Read & write on all entities	Pricing manager
priceprovider.admin:Reader	Read-only access	Auditor
priceprovider.public:PriceRowReader	Public API, org-scoped prices	Customer / shop

Start building with PriceProvider today

Clone the repo · Run one script · Own your pricing engine

github.com/commerce-stack-solutions/priceprovider · commerce-stack-solutions.io

PriceProvider is open-source composable commerce software by commerce-stack-solutions.io · Built on Java 21 / Spring Boot 3 /
Angular 20 · Licensed under open source terms